

Open-World Semantic Segmentation Including Class Similarity

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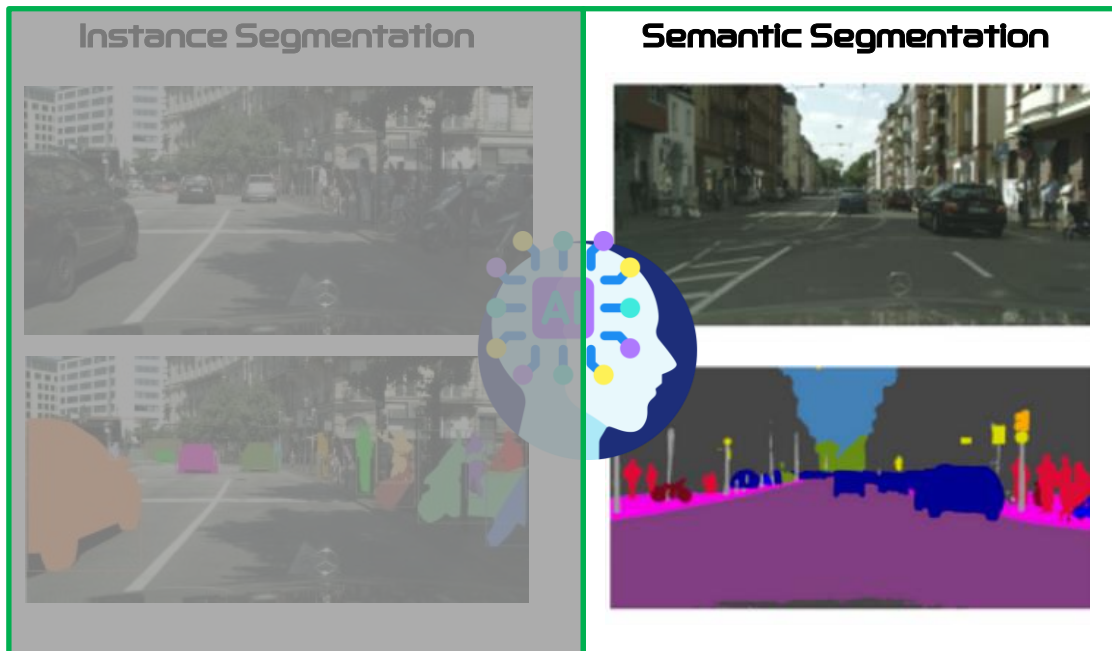
05 | Conclusion

01

Introduction

01. Introduction

1. Open-World Semantic Segmentation



01. Introduction

1. Open-World Semantic Segmentation

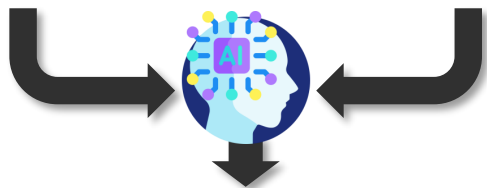
Supervised Learning



[Train Data]



[Annotations]



01. Introduction

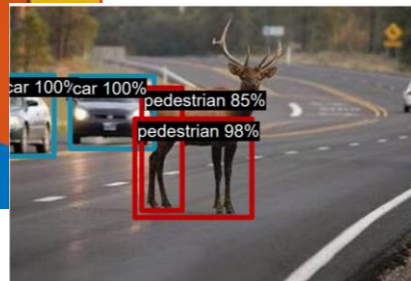
1. Open-World Semantic Segmentation



Good Perception Model

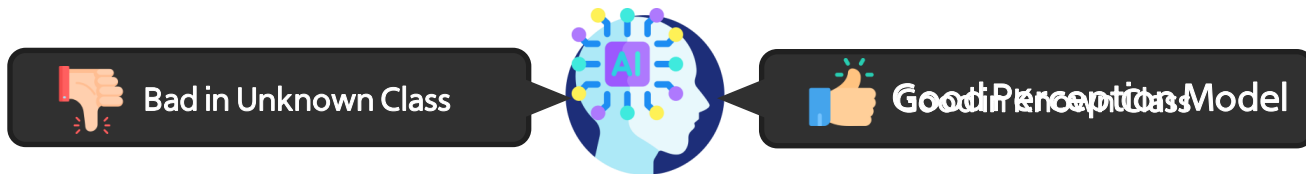


Overconfidence



01. Introduction

1. Open-World Semantic Segmentation



Open-World Semantic Segmentation

01. Introduction

2. ContMAV

1 Fully CNN Based SOTA Network Architecture

- Anomaly Segmentation (w/ Closed-world Semantic Segmentation)

2 Loss Function

- Classify Unknown Classes

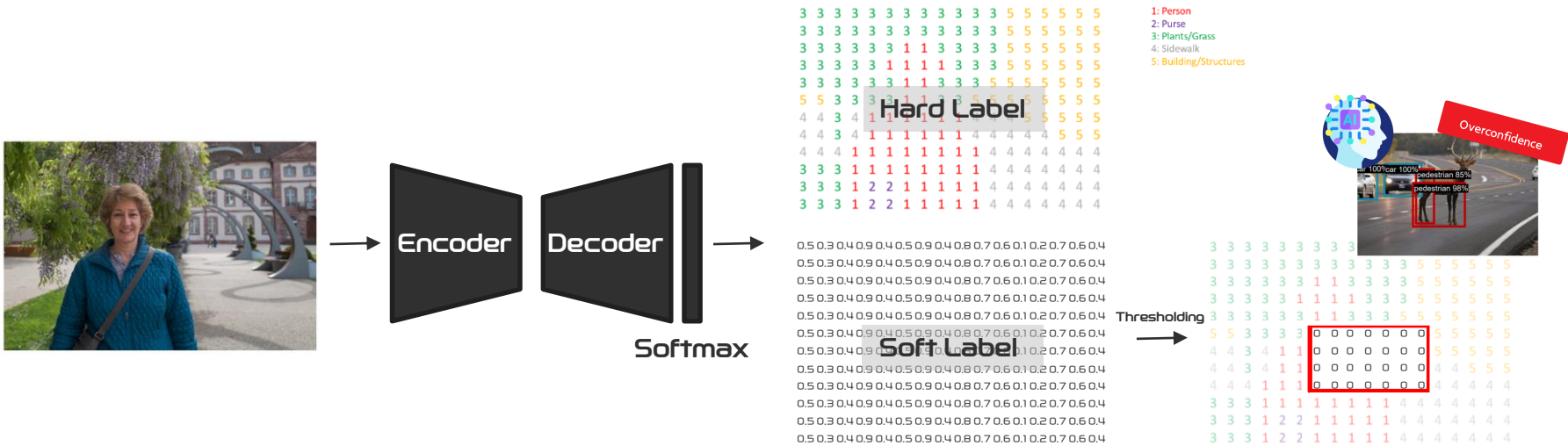
3 Extensive Experiments

02

Related Works

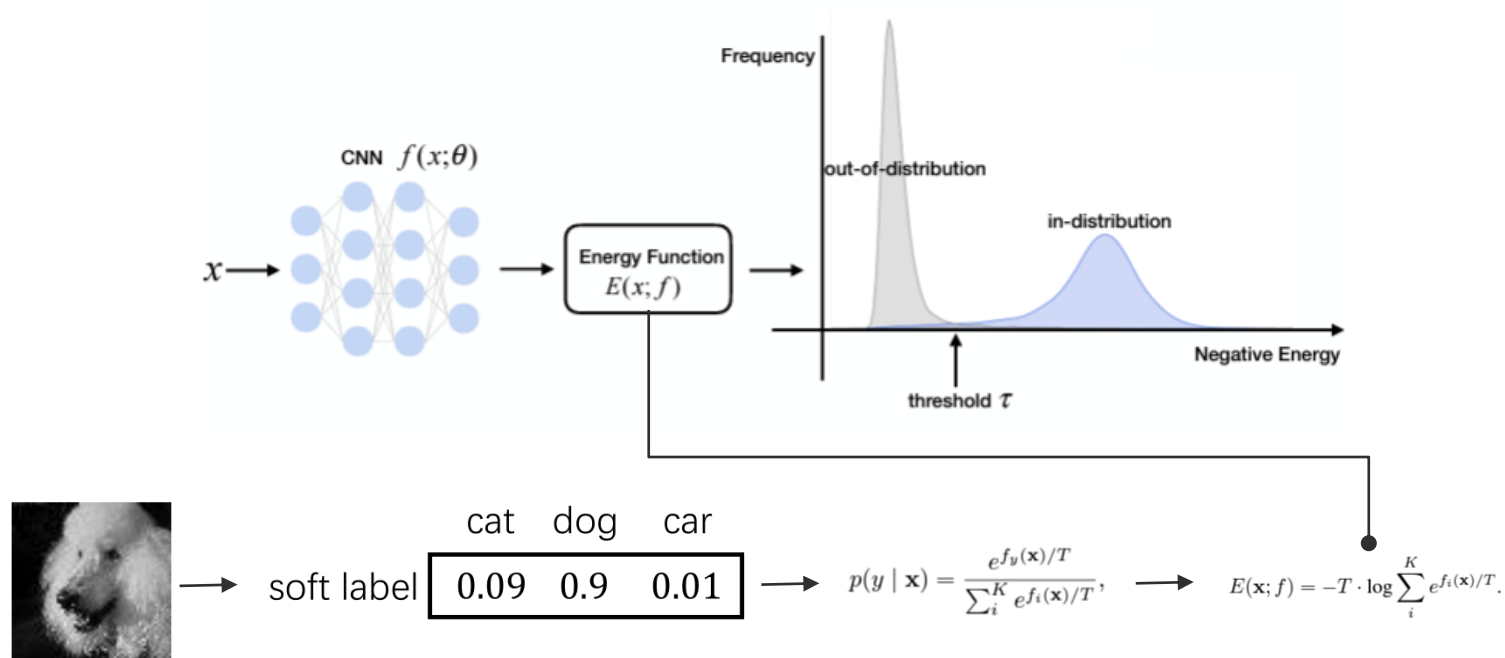
02. Related Works

1. Anomaly Detection and Classification



02. Related Works

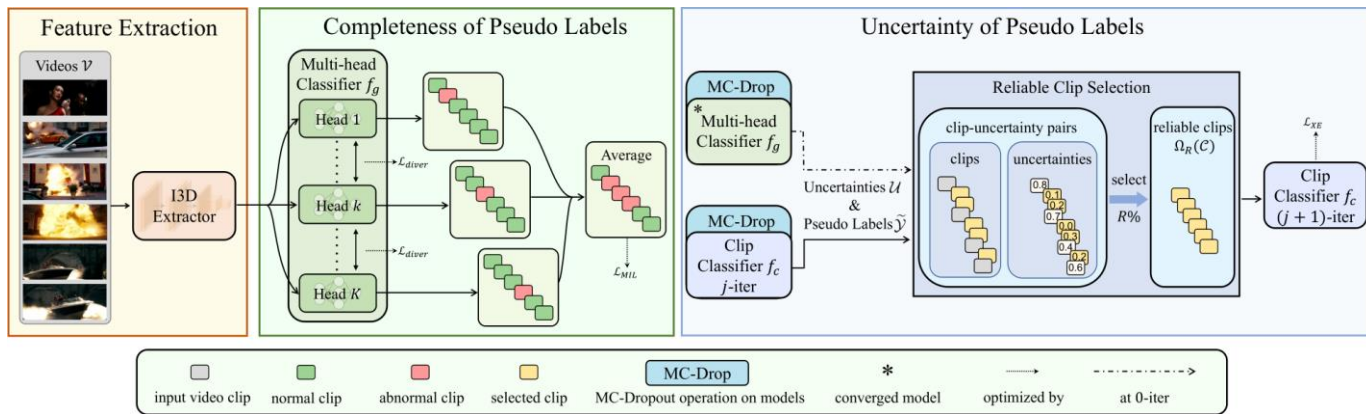
1. Anomaly Detection and Classification



02. Related Works

2. Open-World Segmentation

- Anomaly Detection Based



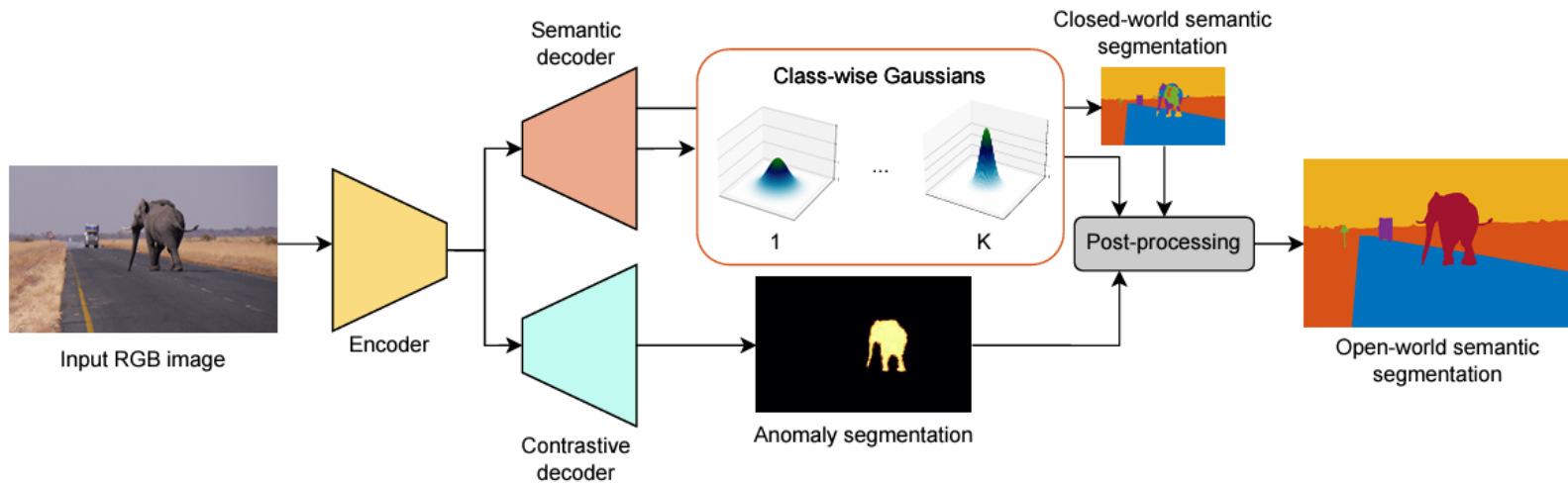
03

Methods

03. Methods

1. General Network Architecture

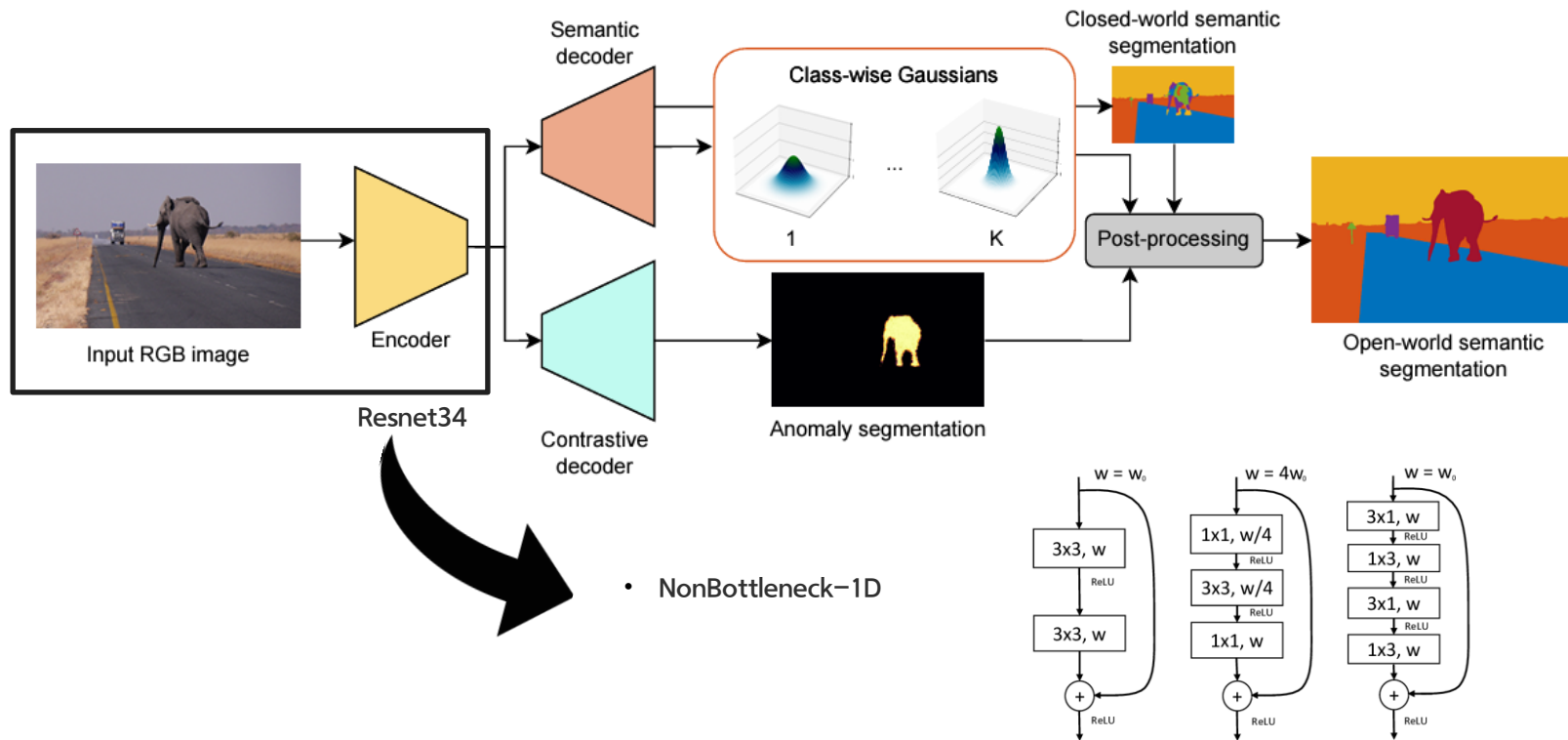
- Architecture



03. Methods

1. General Network Architecture

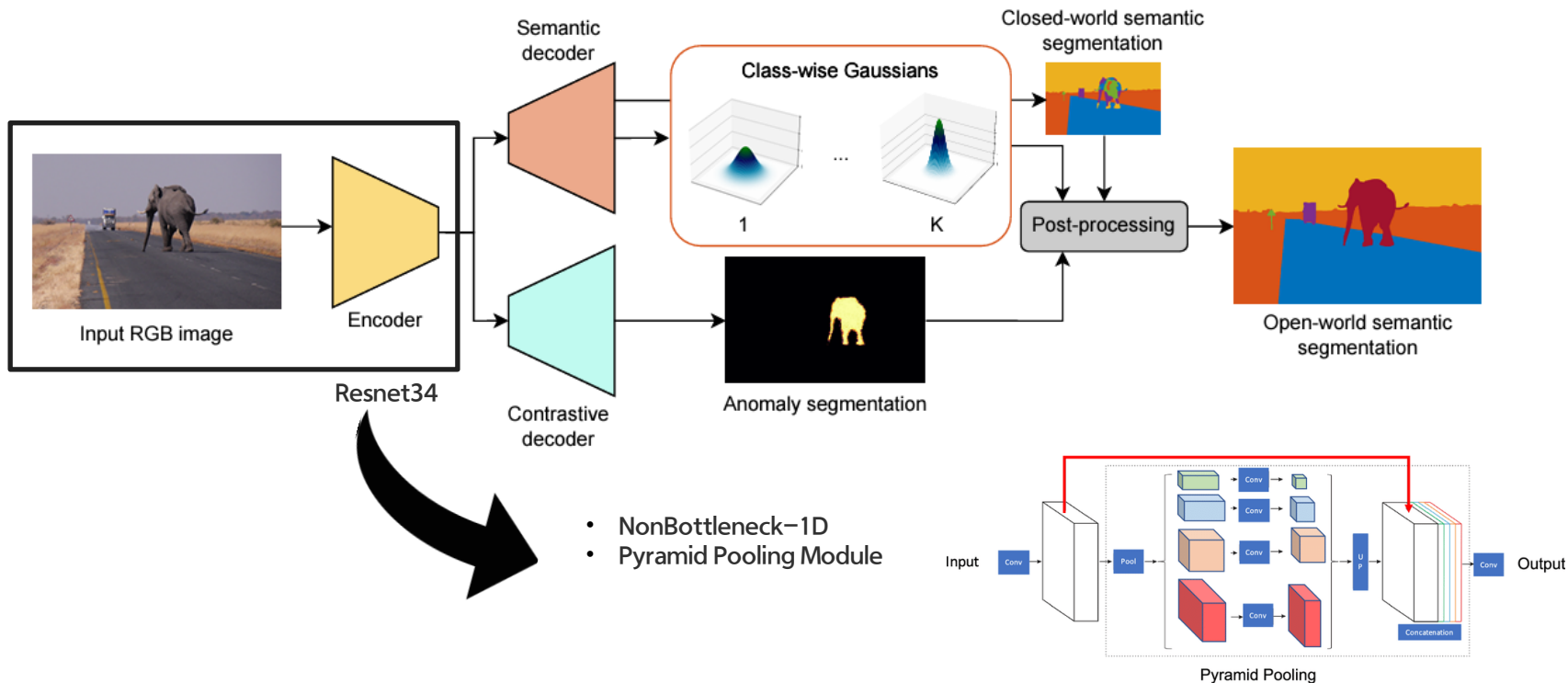
- Image Encoder



03. Methods

1. General Network Architecture

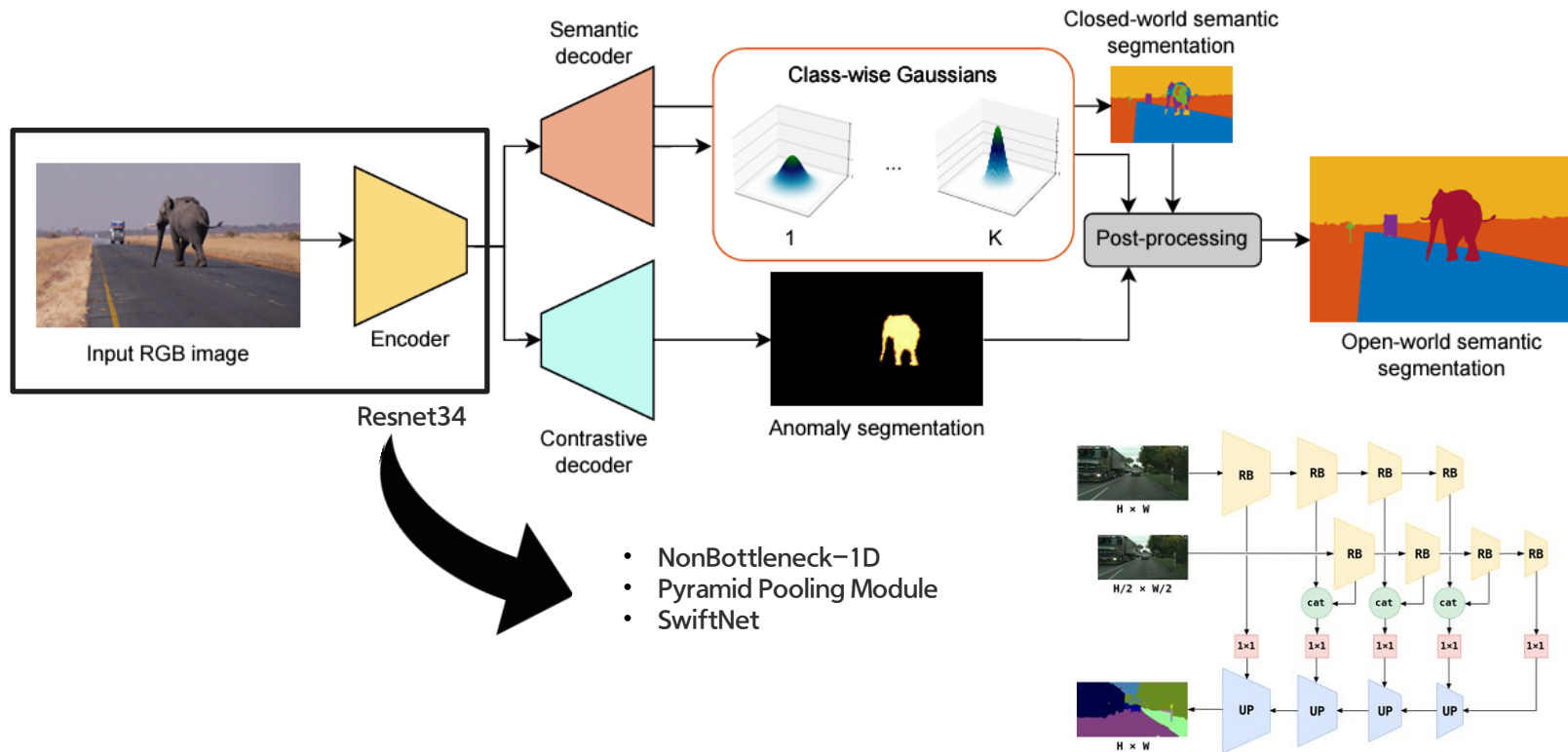
- Image Encoder



03. Methods

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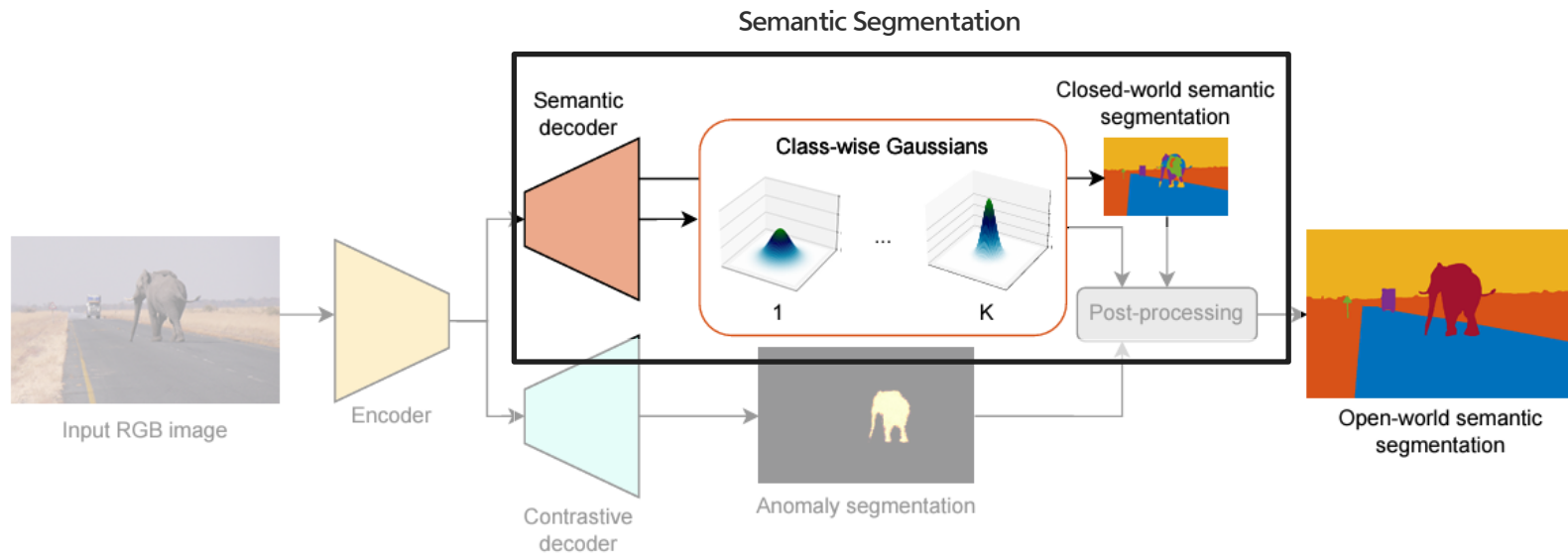
- Image Encoder



03. Methods

2. Approach for Open-World Segmentation

- 1st Decoder : Semantic Decoder



03. Methods

2. Approach for Open-World Segmentation

- 1st Decoder : Semantic Decoder

$$\mathcal{L}_{\text{sdec}} = \underbrace{w_1}_{0.9} \mathcal{L}_{\text{sem}} + \underbrace{w_2}_{0.1} \mathcal{L}_{\text{feat}}.$$

Close-world Semantic Segmentation 성능 유지

Cross Entropy Loss

$$\mathcal{L}_{\text{sem}} = -\frac{1}{|\Omega|} \sum_{p \in \Omega} \omega_k \mathbf{t}_p^\top \log(\sigma(\mathbf{f}_p)),$$

Pre-softmax Feature, Class prototype, Mean Activation Vector

Ω : Pixels $\mathbf{t}$: Ground Truth Logits

ω : Class Weight σ : Softmax Function

Approach	mIoU [%] ↑		
	Train	Motorcycle	Bicycle
Background + cluster	0	32.3	32.8
ContMAV (no feat loss)	48.1	53.8	39.9
ContMAV (with feat loss)	62.4	62.2	56.8
Closed-world	72.3	69.3	60.9

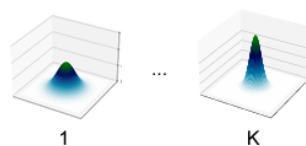
Unique한 Class Descriptor를 가지도록

$$\mathcal{L}_{\text{feat}} = \frac{1}{|\Omega|} \sum_{k=1}^K \sum_{p \in \Omega_k} \frac{\|\mathbf{f}_p - \boldsymbol{\mu}_k^{e-1}\|}{\sigma_k^{e-1}}.$$

$$\boldsymbol{\mu}_k = \frac{1}{|\hat{\Omega}_k|} \sum_{p \in \hat{\Omega}_k} \mathbf{f}_p. \longrightarrow \text{Class Descriptor}$$

$$\mathcal{L}_{\text{feat}} = \frac{1}{|\Omega|} \sum_{k=1}^K \sum_{p \in \Omega_k} \frac{\|\mathbf{f}_p - \boldsymbol{\mu}_k^{e-1}\|}{\sigma_k^{e-1}}.$$

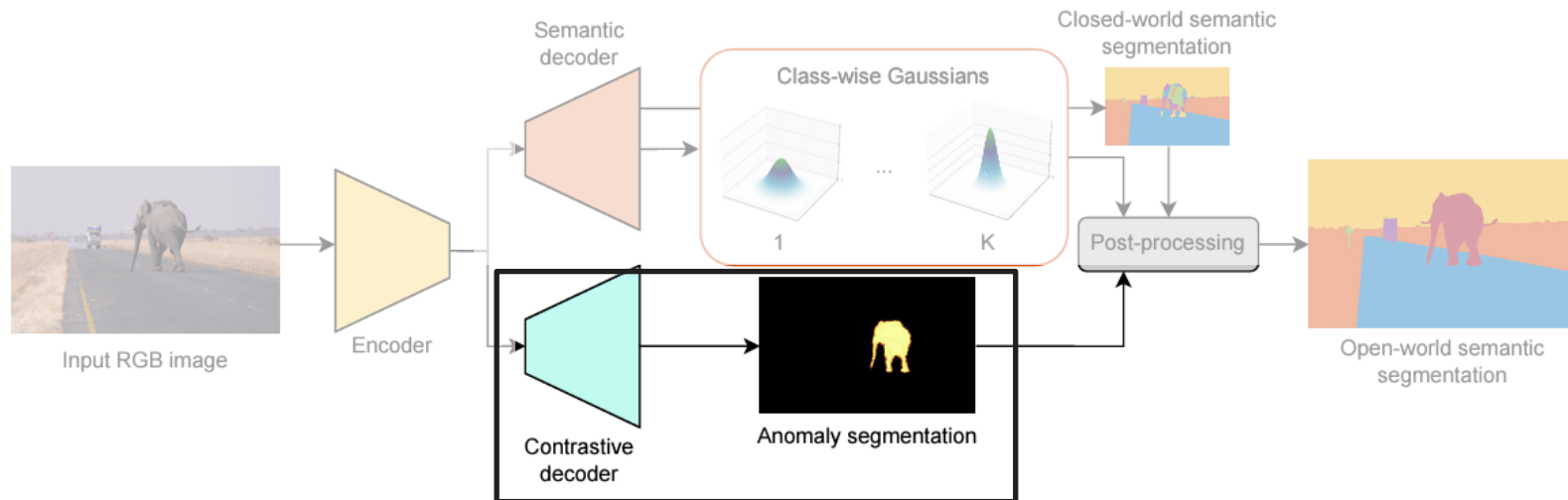
Class-wise Gaussians



03. Methods

2. Approach for Open-World Segmentation

- 2nd Decoder : Contrastive Decoder



03. Methods

2. Approach for Open-World Segmentation

- 2nd Decoder : Contrastive Decoder

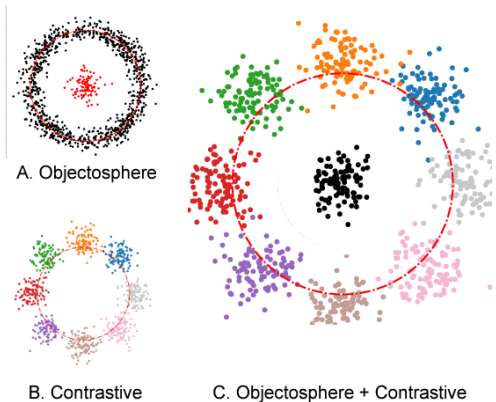
$$\mathcal{L}_{\text{cdec}} = \frac{w_3}{0.5} \mathcal{L}_{\text{cont}} + \frac{w_4}{0.5} \mathcal{L}_{\text{obj}}.$$

다른 Class끼리는 Class Descriptor가 달라지도록

Log like Softmax

$$\mathcal{L}_{\text{cont}} = - \sum_{k=1}^K \log \frac{\exp(\bar{\mathbf{f}}_k^\top \bar{\boldsymbol{\mu}}_k^{e-1} / \tau)}{\sum_{i=1}^K \exp(\bar{\mathbf{f}}_k^\top \bar{\boldsymbol{\mu}}_i^{e-1} / \tau)},$$

$$\bar{\mathbf{f}}_k = \frac{1}{|\Omega_k|} \sum_{p \in \Omega_k} \mathbf{f}_p^d, \quad \longrightarrow \quad \text{동일한 Class의 Class Descriptor 합}$$



Known Class와 Unknown Class 구별 능력 향상

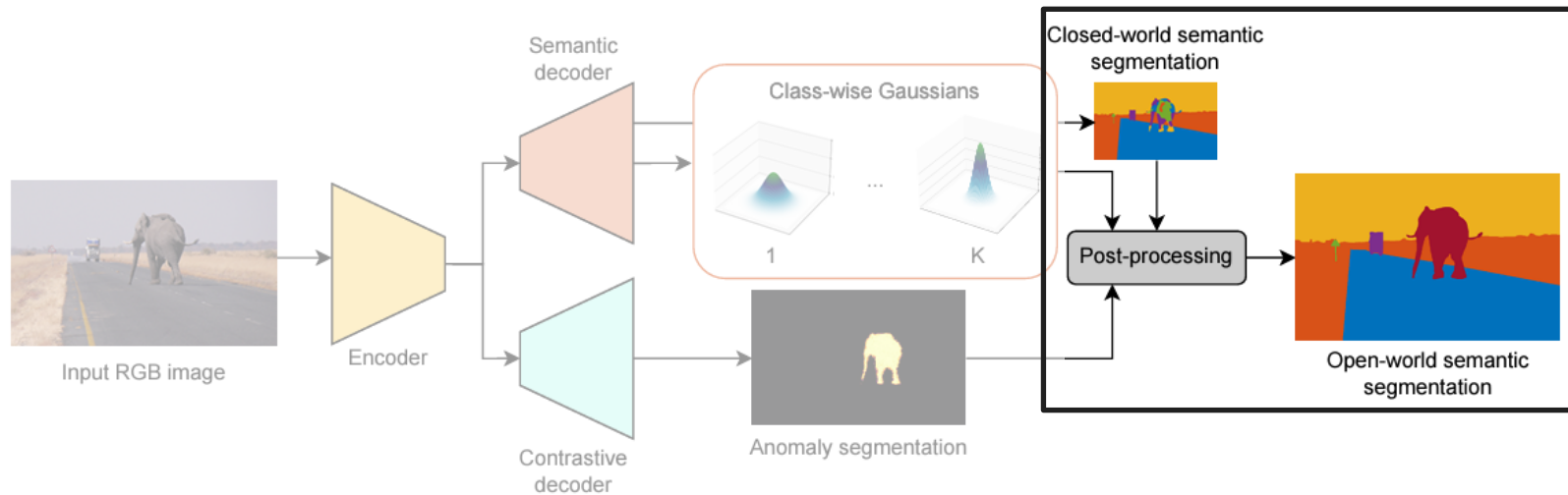
$$\mathcal{L}_{\text{obj}} = \begin{cases} \max(\xi - \|\mathbf{f}_p\|^2, 0) & , \text{if } p \in \mathcal{D}_k \\ \|\mathbf{f}_p\|^2 & , \text{otherwise} \end{cases}$$

→ Known Class는 반지름 ξ Hypersphere 밖으로
Unknown Class는 안으로

03. Methods

2. Approach for Open-World Segmentation

- Post-Processing

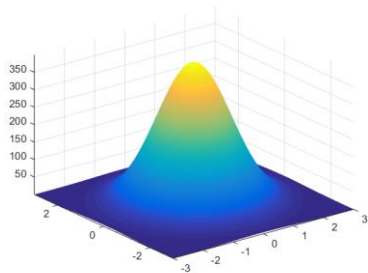


03. Methods

2. Approach for Open-World Segmentation

- Post-Processing

Multi-variate Normal Distribution



$$\mathcal{N}(\mu_k, \Sigma_k), \leftarrow \Sigma_k = \text{diag}(\sigma_k^2)$$

↓ SE Kernel (Sequenced Exponential Kernel) from Gaussian Model

$$s_k(f_p) = \exp\left(-\frac{1}{2}(f_p - \mu_k)^\top \Sigma_k^{-1}(f_p - \mu_k)\right).$$



$$s(p) = \max_k s_k(f_p),$$



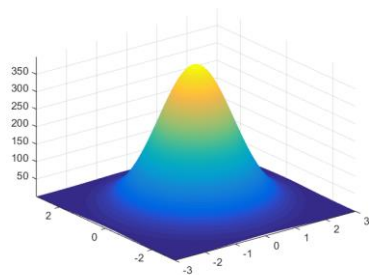
New Class ! (from Semantic Decoder)

03. Methods

Multi-variate Normal Distribution

2. Approach for Open-World Segmentation

- Post-Processing



$$\mathcal{N}(\mu_k, \Sigma_k), \leftarrow \Sigma_k = \text{diag}(\sigma_k^2)$$



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$$s_k(f_p) = \exp\left(-\frac{1}{2}(f_p - \mu_k)^\top \Sigma_k^{-1}(f_p - \mu_k)\right).$$



$$s(p) = \max_k s_k(f_p), \quad + \quad s_{\text{unk}, p}^{\text{cont}} = \max\left(0, \left(1 - \frac{\|f_p\|^2}{\xi}\right)\right),$$

New Class ! (from Semantic Decoder)

Unknown Class (from Contrastive Decoder)



$$s_{\text{unk}, p} = \frac{1}{2} \left(s_{\text{unk}, p}^{\text{sem}} + s_{\text{unk}, p}^{\text{cont}} \right).$$

Q & A

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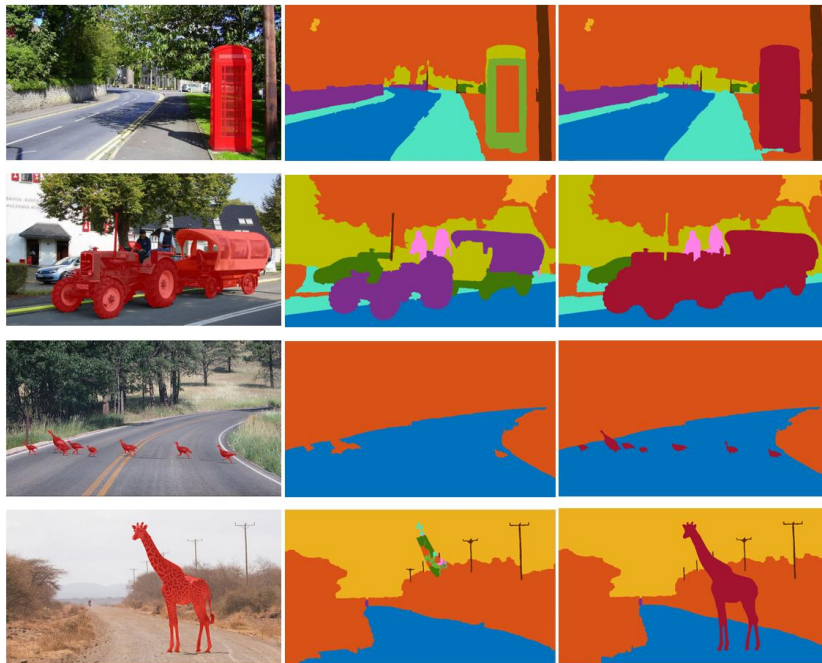
04

Experiments

04. Experiments

1. Datasets

SegmentMelfYouCan

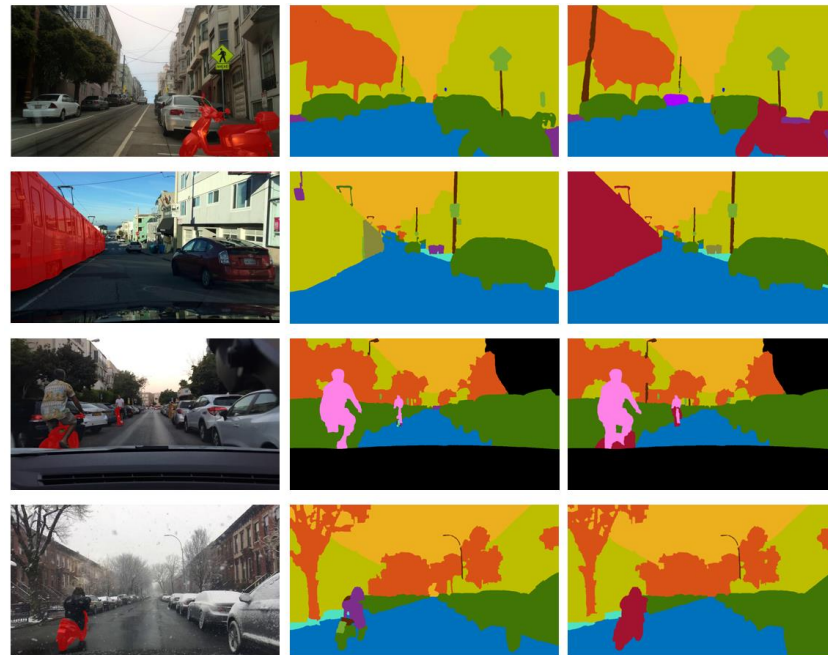


(a) Input RGB

(b) Closed-world prediction

(c) Open-world prediction

BDDAnomaly



(a) Input RGB

(b) Closed-world prediction

(c) Open-world prediction

04. Experiments

2. Anomaly Segmentation

Approach	mIoU [%] ↑	
	CityScapes	BDDAnomaly
CW	71.1	64.1
OW	70.8	62.8

Close World와 Open World 성능차이 거의 없음

Approach	OoD	Pixel-Level		Component-Level		
		AUPR [%] ↑	FPR95 [%] ↓	sIoU gt [%] ↑	PPV [%] ↑	mean F1 [%] ↑
DenseHybrid [19]	✓	78.0	9.8	54.2	24.1	31.1
RbA [47]	✓	94.5	4.6	64.9	47.5	51.9
Maskomaly [1]	✗	93.4	6.9	55.4	51.2	49.9
RbA [47]	✗	86.1	15.9	56.3	41.4	42.0
ContMAV (ours)	✗	90.2	3.8	54.5	61.9	63.6

Approach	AUPR [%] ↑	FPR95 [%] ↓
MaxSoftmax [23]	3.7	24.5
Background [6]	1.1	40.1
MC Dropout [16]	4.3	16.6
Confidence [14]	3.9	24.5
MaxLogit [24]	5.4	14.0
ContMAV (ours)	96.1	6.9

04. Experiments

2. Anomaly Segmentation

Approach	Accuracy [%] ↑	
	Motorcycle	Train
Baseline	12.5	9.8
ContMAV with MA	39.9	27.6
ContMAV	58.9	49.9



(a) Input RGB

(b) Closed-world prediction

(c) Open-world prediction

Segment Me
IF YOU CAN

Leaderboard

Datasets

Submit

Paper

Slides

Video

Code

LEADERBOARD

Evaluation Metrics

- **AUPR**: pixel-wise Area Under Precision Recall curve
- **FPR₉₅**: pixel-wise False Positive Rate at a true positive rate of 95%
- **sIoU gt**: adjusted Intersection over Union averaged over all ground truth segmentation components
- **PPV**: predictive positive value (or precision) averaged over all predicted segmentation components
- **mean F1**: component-wise F1-score averaged over different detection thresholds

For a more detailed explanation of the metrics, we refer to our [paper](#).

Anomaly Track

Method	OoD Data	Pixel Level		Component Level		
		AUPR ↑	FPR ₉₅ ↓	sIoU gt ↑	PPV ↑	mean F1 ↑
ContMAV		90.20%	3.83%	54.55%	61.86%	63.64%
EAM [paper]	✓	93.75%	4.09%	67.09%	53.77%	60.86%
RbA [paper]	✓	94.46%	4.60%	64.93%	47.51%	51.87%
Maskomaly [paper] [code]	✗	93.35%	6.87%	55.43%	51.46%	49.90%
CSL	✗	80.08%	7.16%	46.46%	50.02%	50.39%
DenseHybrid [paper] [code]	✓	77.96%	9.81%	54.17%	24.13%	31.08%
cDNP [paper] [code]	✗	88.90%	11.42%	50.44%	29.04%	28.12%
RPL+CoroCL [paper] [code]	✓	83.49%	11.68%	49.77%	29.96%	30.16%
Mask2Anomaly [paper] [code]	✓	88.72%	14.63%	55.28%	51.68%	47.16%
Maximized Entropy [paper] [code]	✓	85.47%	15.00%	49.21%	39.51%	28.72%

05

Conclusion

05. Conclusion

Summary



Close-World의 문제를 해결하는 성능 좋은 Open World Semantic Segmentation

- Train - Inference 간의 Feedback Pipeline을 잘 구축하면 자가 학습 기반의 좋은 흐름이 만들어 질 것 같음



많은 Hand Crafted Hyper Parameter

Architecture를 선택한 과정에 대한 설명



데이터셋 구성에 따른 성능 차이

왜 Unknown Class들이 잘 구별되는 것인가?

- 손실 함수 기중치

◦ $w_1: 0.9$

◦ $w_2: 0.1$

◦ $w_3: 0.5$

◦ $w_4: 0.5$

The semantic decoder is thus optimized with a weighted sum of the loss functions introduced above

$$\mathcal{L}_{sdec} = w_1 \mathcal{L}_{sem} + w_2 \mathcal{L}_{feat}. \quad (6)$$

The contrastive decoder is optimized with a weighted sum of the two losses given by

$$\mathcal{L}_{cdec} = w_3 \mathcal{L}_{cont} + w_4 \mathcal{L}_{obj}. \quad (10)$$

주요 하이퍼파라미터

- Contrastive Decoder의 특성 및 이상치 분포를 조정하는 파라미터:

• ϵ ($\epsilon=1$) $\mathcal{L}_{contrastive}$ 손실 함수에 의해 생성되는 $\mathcal{L}_{perseg}$ 빈자들

Table 7. Anomaly segmentation results on BDDAnomaly with different choices of the parameter ϵ .

Approach	AUFR [%] ↑	FPR95 [%] ↓
ContMAV ($\epsilon = 0.75$)	92.2	18.7
ContMAV ($\epsilon = 1.25$)	83.4	55.2
ContMAV ($\epsilon = 1$)	96.1	6.9

- Contrastive 손실과 함께 모든 특징 벡터를 단위 구에 분포시키도록 설계.

• 이로 인해 δ 값이 1이 아닌 경우 성능이 저하될 수 있음

- Threshold δ (delta) 0.6 ("unknown **new**") 임계값으로, 픽셀이 새로운 카테고리에게 속하는지 결정하는 데 사용

Table 8. Anomaly segmentation results on BDDAnomaly with different choices of the parameter δ .

Approach	AUFR [%] ↑	FPR95 [%] ↓
ContMAV ($\delta = 0.4$)	86.6	41.2
ContMAV ($\delta = 0.8$)	89.1	30.1
ContMAV ($\delta = 0.6$)	96.1	6.9

- 점수는 semantic 및 contrastive 디코더에서 생성된 점수의 평균으로 계산

• 낮은 값은 많은 false positives를, 높은 값은 많은 false negatives를 발생

- τ (tau) 0.1 (Contrastive Loss의 온도 매개변수, 클래스 간의 유사성을 조절)

• Chen et al. (2020)의 제안에 따름

- η (eta) 0.5 (Mean Activation Vector의 유사도 임계값)

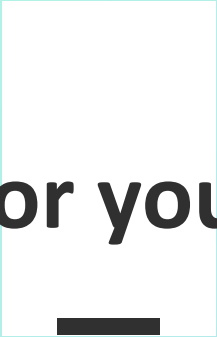
• 픽셀이 새로운 클래스인지 또는 이미 발견된 새로운 클래스 중 하나에 속하는지를 결정하는 최소 거리로, 데이터 분포에 따라 최적의 값을 달리함

Table 9. Class discovery results on BDDAnomaly with different choices of the parameter η . For each class of interest, the discovered one with greater IoU is chosen and reported.

Approach	Train	mIoU [%] ↑		N_U
		Motorcycle	Bicycle	
ContMAV ($\eta = 0.3$)	0.0	23.4	0.0	1
ContMAV ($\eta = 0.9$)	30.5	31.1	18.9	12
ContMAV ($\eta = 0.6$)	62.4	62.2	56.8	4

Q & A

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Thank you for your attention